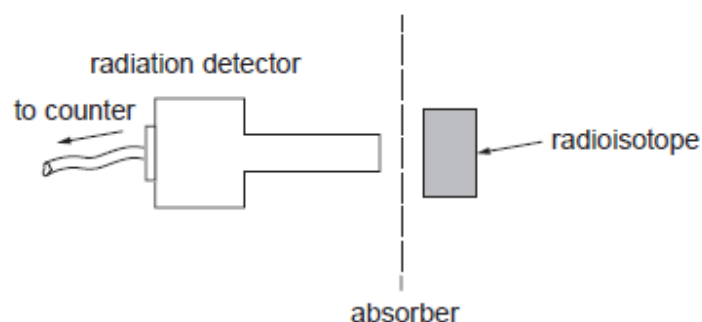


#1

8. The radiation emitted by radioisotopes can be investigated using the apparatus in the diagram below. A radioisotope was placed 1 cm from a radiation detector. Different absorbers were placed, one at a time, between the radioisotopes and the detector and the count rate measured.



The following table shows how the count rate for four different radioisotopes is affected by different absorbers.

Radioisotope	Count rate for different absorbers (counts per second)			
	air	paper	aluminum	lead
sulfur-35	200	199	58	25
radium-223	200	5	5	5
cobalt-57	250	141	140	20
strontium-90	220	221	5	5

Use the information in the table opposite to answer the questions below.

- (a) Estimate the value of the background radiation. [1]

..... counts per second

- (b) (i) Explain what conclusions can be drawn about the types of radiation emitted by cobalt-57. [3]

.....

.....

.....

.....

.....

- (ii) It is suggested that radium-223 does not need to be stored in a metal box. Explain whether you agree with this suggestion. [3]

.....

.....

.....

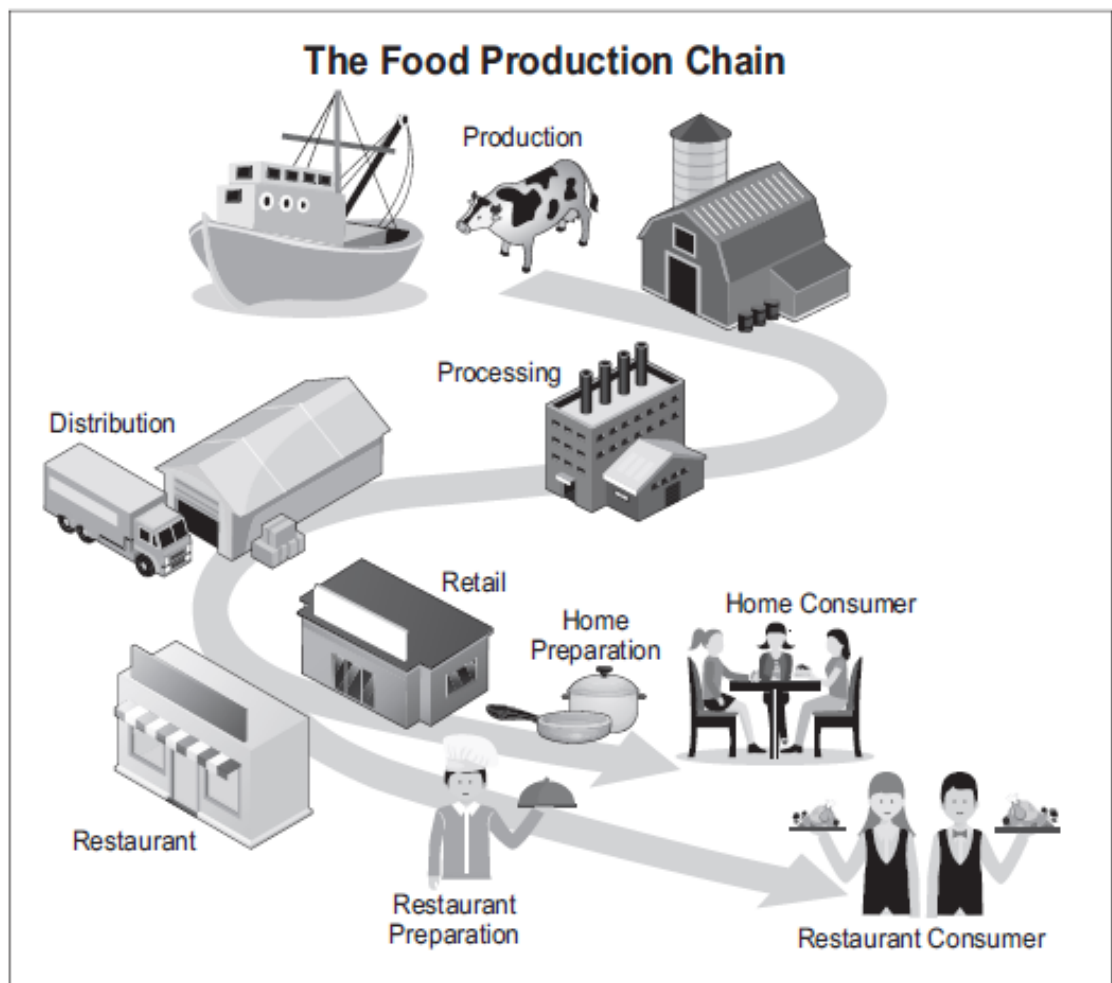
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Question taken from WJEC examination paper 344201, September 2019

2. The food industry uses fruits, vegetables, grains and raw meat and processes these into the food products that are sold in supermarkets. Food production often involves cleaning and cooking raw foods, as well as the addition of additives and other ingredients.

One purpose of food production processes is to create products with longer shelf-lives than raw food ingredients.



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The food industry must take precautions to prevent consumers becoming ill from food poisoning.

- (a) Complete the table below which shows some of the methods of controlling bacteria in the food industry. [3]

Method	Effect on bacteria
refrigeration	slows down bacterial growth
freezing	
	kills bacteria
	dries bacteria to prevent growth

(b) Sometimes food becomes contaminated with bacteria, causing food poisoning.

(i) State one symptom of food poisoning. [1]

.....

(ii) Name one type of bacteria that causes food poisoning. [1]

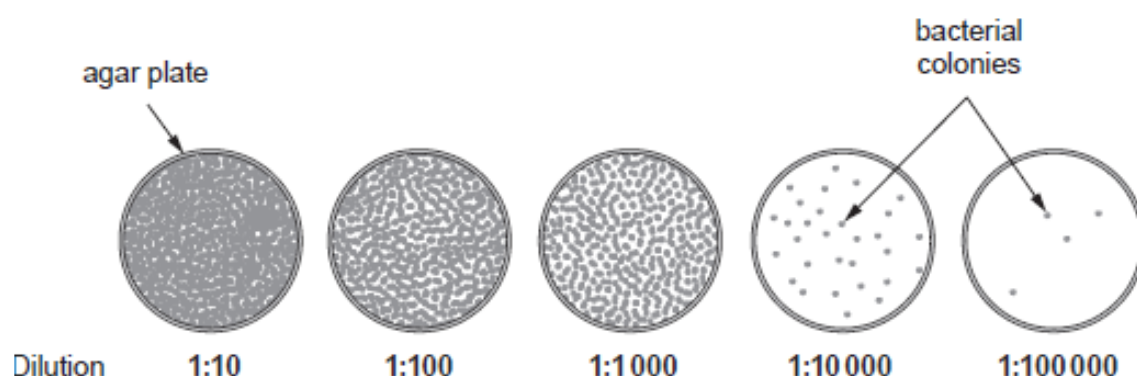
.....

(iii) State the harmful substances produced by bacteria as they grow. [1]

.....

(c) Whenever an outbreak of food poisoning occurs, environmental health officers test samples of food for the presence of bacteria.

(i) To be able to count the number of bacterial colonies a series of dilutions is carried out as shown in the diagram below.



The 1:10000 dilution was used to calculate the number of bacteria present.

I Explain why using a dilution of 1:10 will not provide valid results. [2]

.....
.....
.....

II Give one reason why the 1:100000 dilution was also not used to calculate the number of bacteria present. [1]

.....

- (ii) In an investigation, six agar plates were prepared by adding 1 cm^3 of solution from a 1:10 000 dilution. The agar plates were incubated for 72 hours. The results are shown below.

Plate number	1	2	3	4	5	6
Number of colonies	35	35	25	15	30	25

- I State which plate has produced an anomalous result. [1]

.....

- II Use the equation:

$$\text{number of bacteria/cm}^3 = \text{mean number of colonies} \times 10\,000$$

- to calculate the number of bacteria/cm³ in the original sample. [3]

$$\text{Number of bacteria/cm}^3 = \dots\dots\dots$$

A food manufacturer claims that probiotic yoghurt, containing live bacteria, will provide health benefits when eaten.

(a) Describe the stages in the production of yoghurt. [3]

.....

.....

.....

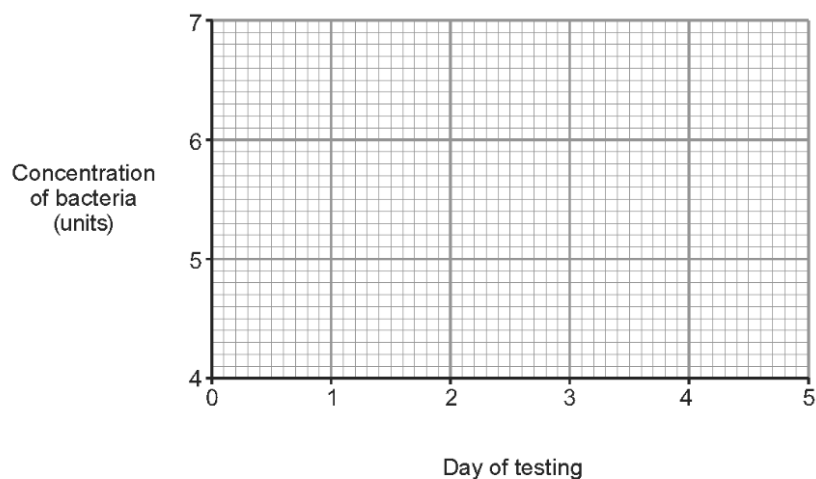
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(b) Some people claim that bacteria will not survive in the stomach. The food manufacturer claims that the bacteria will survive and their numbers will increase.

An independent scientist investigates the claim. She produces a model of the stomach and adds live bacteria found in the yoghurt. The bacteria concentration is measured daily for 5 days. The results are shown below.

Day of testing	Concentration of bacteria (units)
1	5.0
2	6.4
3	4.8
4	5.6
5	4.8

- (i) Plot a graph of the data opposite on the grid below and join the plots, point to point. [3]



- (ii) Do the results agree with the claim that the bacteria will **not** survive in the stomach? Give **one** reason for your answer. [1]

.....

.....

- (iii) Do the results agree with the manufacturer's claim that the number of bacteria in the stomach will increase? Give **one** reason for your answer. [1]

.....

.....

- (c) (i) It is important that harmful bacteria do not get into the probiotic yoghurt during production.

State **two** methods of making sure this will not happen. [2]

1.
2.

- (ii) Name **one** symptom of food poisoning caused by harmful bacteria. [1]

.....

Intensive farming methods include the use of pesticides, fertilisers, herbicides, and 'battery' farming. These methods increase food production but there are disadvantages.

Complete the table using words from the box below.
The first line has been completed for you. Do not use a word more than once. [6]

input	nutrients	plants	output	biodiversity	indoors	rivers
insects	warm	sunlight	<u>animals</u>	<u>poison</u>		

Intensive farming method	Reason for use	Disadvantage
Pesticide spray	Removes <u>animals</u> that feed on the crop	May <u>poison</u> useful organisms
Herbicide spray	Removes competing	Reduces
Fertiliser spray	Adds to the soil	Causes dense plant growth in
'Battery' farming	Increases	Keeps animals

Marking Scheme

#1

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			5		1		1		
	(b)	(i)		Drop with paper therefore contains α (1) Aluminium makes no difference so no β (1) Some radiation passes through lead so γ / aluminium makes no difference so γ (1)			3	3		
		(ii)		(All) radiation is absorbed by paper (1) So only α (1) Won't travel far through air (1)			3	3		6
				Question 8 total	0	1	6	7	0	6

#2

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2 FT	(a)			stops bacterial growth (1) cooking/heating/boiling (1) salting (1)	3			3		
	(b)	(i)		Any 1 $\times$ (1) from: Vomiting (1) diarrhoea (1) stomach pains /nausea (1)	1			1		
		(ii)		<i>Campylobacter</i> / <i>E. coli</i> / <i>Salmonella</i> (1)	1			1		
		(iii)		Toxins / poisons (1)	1			1		
	(c)	(i)	I	colonies have clumped together (1) Unable to count individual colonies/too many colonies to count (1)			2	2		2
			II	Not enough colonies (for valid results) (1)			1	1		1
		(ii)	I	Plate 4 Accept 15			1	1		1
			II	Mean number of colonies = $\frac{150}{5}$ (ecf) (1) = 30 (1) (Allow sum of remaining 5 values as ecf from previous answer) Count = 30 (ecf) $\times$ 10 000 = 300 000 / cm ³ (1) subs Or Mean number of colonies = $\frac{165}{6}$ = 27.5 (1) Count = 27.5 (ecf) $\times$ 10 000 = 275 000 / cm ³ (1) subs Accept 280 000 Allow (1) if candidate has ignored anomaly but calculated the answer incorrectly	1	2		3	3	3
				Question 2 total 13	7	2	4	13	4	7

#3

Question	Marking Point	Mark
5 (a)	Heat the milk (to sterilise it), add the culture (bacteria), store at about 40 ⁰ C for a few hours.	3
(b) (i)	plots (2) dot to dot line(1)	3
(ii)	(No) still bacteria present after 5 days	1
(iii)	(No) still the same after 5 days (within a tolerance)	1
(c) (i)	Sterilise equipment / personal hygiene / avoid cross contamination	2
(ii)	sickness / diarrhoea	1

#4

Question	Marking detail	Mark												
2	1 mark for each correct answer <i>Correct words (bold text) to complete statement are shown in the table below.</i> <table> <tr> <th>Intensive farming method</th><th>Reason for use</th><th>Disadvantage</th></tr> <tr> <td>Herbicide spray</td><td>plants</td><td>biodiversity</td></tr> <tr> <td>Fertiliser spray</td><td>nutrients</td><td>rivers</td></tr> <tr> <td>'Battery' farming</td><td>output</td><td>indoors</td></tr> </table>	Intensive farming method	Reason for use	Disadvantage	Herbicide spray	plants	biodiversity	Fertiliser spray	nutrients	rivers	'Battery' farming	output	indoors	6
Intensive farming method	Reason for use	Disadvantage												
Herbicide spray	plants	biodiversity												
Fertiliser spray	nutrients	rivers												
'Battery' farming	output	indoors												